

# Dynamic Activation Quantization

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# Flip Rates

4-bit quantization of Dream-7B-Base and LLaDA-8B-Base. The first/second number indicates the % of correct/incorrect answers of the baseline model that changed.

| Model                | ARC-Challenge | ARC-Easy    | PIQA        | Winogrande  | MMLU        | HellaSwag   |
|----------------------|---------------|-------------|-------------|-------------|-------------|-------------|
| LLaDA-8B-Base (SINQ) | 1.88 / 1.28   | 1.60 / 1.30 | 1.03 / 0.49 | 2.84 / 2.84 | 3.85 / 4.37 | 0.64 / 0.28 |
| LLaDA-8B-Base (AWQ)  | 2.90 / 0.85   | 1.60 / 1.64 | 0.82 / 0.76 | 3.24 / 2.84 | 4.56 / 4.24 |             |
| LLaDA-8B-Base (GPTQ) | 2.65 / 2.05   |             | 1.41 / 1.09 | 3.08 / 3.16 |             |             |
| Dream-7B-Base (SINQ) | 3.67 / 2.56   | 2.06 / 1.35 | 1.85 / 1.52 | 3.39 / 2.29 | —           | —           |
| Dream-7B-Base (AWQ)  | 4.10 / 2.13   | 2.06 / 1.18 | 1.47 / 0.92 | 3.08 / 2.05 | —           | —           |

# DLMs are Compute Bound

- DLMs throughput is significantly lower than AR and block diffusion models
- DLMs require encoding the entire sequence when modeling each token, which increases the computation cost

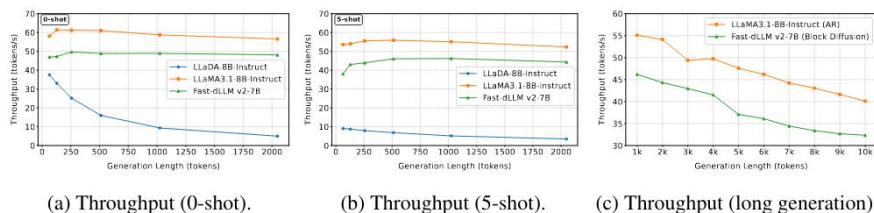


Figure 1: (a) and (b) show the throughput comparison across different generation lengths under the 0-shot and 5-shot settings, respectively, while (c) presents the throughput comparison between AR and block diffusion models under longer generation lengths.

# 4-bit Dynamic Quantization of SINQ Llada

- Used absmax quantization for quantizing activations during inference.
- Dequantize the activations before the matmul operation

```
# per-token: find the largest absolute value in each token (row)
scale_x = x_scaled.abs().amax(dim=-1, keepdim=True).clamp(min=1e-8) / 7.0
# divide by scale so values fit in [-7, 7], round to integers, clip to INT4 range
x_q = (x_scaled / scale_x).round().clamp(-8, 7)
# multiply back by scale to return to fp16 - values are now snapped to INT4 grid
x_scaled = x_q.to(x.dtype) * scale_x
```

```
(Pdb) x_q
tensor([[[[-0.,  0., -0., ..., -0., -0., -1.],
          [ 0., -0., -0., ..., -0.,  0.,  0.],
          [ 0.,  0.,  0., ..., -0.,  0.,  0.],
          ...,
          [ 0., -0.,  0., ...,  0., -0.,  0.],
          [ 0., -0.,  0., ...,  0., -0.,  0.],
          [ 0., -0.,  0., ...,  0., -0.,  0.]]], device='cuda:0',
        dtype=torch.bfloat16)
(Pdb) x_q.unique()
tensor([-7., -6., -5., -4., -3., -2., -1., -0.,  1.,  2.,  3.,  4.,  5.,  6.,
         7.], device='cuda:0', dtype=torch.bfloat16)
(Pdb) x_q.float().mean()
tensor(0.0005, device='cuda:0')
(Pdb) x_q.float().std()
tensor(0.1581, device='cuda:0')
```

```
(Pdb) x_dq = x_q.to(self.compute_dtype) * scale_x
(Pdb) (x - x_dq).abs().max()
tensor(26.5000, device='cuda:0', dtype=torch.bfloat16)
```

# 8-bit Dynamic Quantization of SINQ Llada

**Prompt: Joy can read 8 pages of a book in 20 minutes. How many hours will it take her to read 120 pages?**

Response: Joy can read 8 pages in 20 minutes, which is equivalent to 8 pages per hour (since there are 60 minutes in an hour). To find out how many hours it will take her to read 120 pages, we divide the total number of pages by the number of pages she can read in an hour:  $120 \text{ pages} / 8 \text{ pages per hour} = 15 \text{ hours}$  Therefore, it will take Joy 15 hours to read 120 pages. ##### 15 The answer is: 15 ##### 15 The answer

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**Prompt: Randy has 60 mango trees on his farm. He also has 5 less than half as many coconut trees as mango trees. How many trees does Randy have in all on his farm?**

Response: To find the total number of trees on Randy's farm, we need to add the number of mango trees and the number of coconut trees. First, let's find the number of coconut trees. Randy has 5 less than half as many coconut trees as mango trees. Half as many coconut trees as mango trees would be  $60 / 2 = 30$  coconut trees. So, Randy has  $30 - 5 = 25$  coconut trees. Now, let's add the number of mango trees and coconut trees to find the total number of trees on Randy's farm.

# W4A16 Outputs

**Prompt: Joy can read 8 pages of a book in 20 minutes. How many hours will it take her to read 120 pages?**

Joy can read 8 pages in 20 minutes, which is equivalent to 8 pages per hour (since there are 60 minutes in an hour and 20 minutes is half of an hour). To find out how many hours it will take her to read 120 pages, we divide the total number of pages by the number of pages she can read per hour:  $120 \text{ pages} / 8 \text{ pages per hour} = 15 \text{ hours}$  So, it will take Joy 15 hours to read 120 pages of the book.

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**Prompt: Randy has 60 mango trees on his farm. He also has 5 less than half as many coconut trees as mango trees. How many trees does Randy have in all on his farm?**

First, let's find out how many coconut trees Randy has. Since he has 5 less than half as many coconut trees as mango trees, we can calculate this as follows:  $\text{Half as many coconut trees as mango trees} = 60 / 2 = 30$  coconut trees  $5 \text{ less than half as many coconut trees as mango trees} = 30 - 5 = 25$  coconut trees So, Randy has 25 coconut trees. Now, to find out the total number of trees Randy has on his farm, we add the number of mango trees to the number of coconut trees:

# Mask-Conditioned Activations Experiment

- The goal is to find a relationship between mask ratios and activations to predict proper scaling factors for activation quantization
- Test whether a weighted average of two reference states can predict the activations at each layer

$$h^{(\ell)}(r) \approx r \cdot A^{(\ell)} + (1 - r) \cdot B^{(\ell)}$$

- $r$ : mask ratio and is used as weighting factor
- $A$ : activation values captured when the model input is fully masked ( $r=1$ )
- $B$ : activation values captured when the model input is fully unmasked ( $r=0$ )

# Mask-Conditioned Activations Experiment

- Run forward passes through llada model with different mask ratios, 0,25%,50%,75% and 100%
- At each of the 32 transformer blocks capture the activations  $f$
- Compare the activations at each layer with the activations predicted using the weighted combination of the two states, compare using
  - Cosine Similarity
  - Relative error



# Mask-Conditioned Activations Results

Table 15: Per-layer linearity check: cosine similarity and relative error between actual hidden states and linear mixture prediction  $r \cdot A^{(\ell)} + (1-r) \cdot B^{(\ell)}$  across mask ratios. **Green** : cosine  $\geq 0.95$  (linear mixture viable). **Yellow** :  $0.80 \leq \text{cosine} < 0.95$  (mild deviation). **Red** : cosine  $< 0.80$  (linear mixture fails).

| Layer | $r = 0.25$ |          | $r = 0.50$ |          | $r = 0.75$ |          |
|-------|------------|----------|------------|----------|------------|----------|
|       | Cosine     | Rel. Err | Cosine     | Rel. Err | Cosine     | Rel. Err |
| 0     | 0.9777     | 0.2039   | 0.9567     | 0.3018   | 0.9235     | 0.3895   |
| 1     | 0.9758     | 0.2254   | 0.9497     | 0.3435   | 0.9143     | 0.4269   |
| 2     | 0.9769     | 0.2319   | 0.9538     | 0.3356   | 0.9222     | 0.4076   |
| 3     | 0.9744     | 0.2584   | 0.9502     | 0.3690   | 0.9229     | 0.4391   |
| 4     | 0.9671     | 0.3003   | 0.9344     | 0.4177   | 0.9027     | 0.5002   |
| 5     | 0.9621     | 0.3187   | 0.9239     | 0.4338   | 0.8881     | 0.5231   |
| 6     | 0.9487     | 0.4133   | 0.8979     | 0.5490   | 0.8516     | 0.6478   |
| 7     | 0.9335     | 0.4493   | 0.8688     | 0.6108   | 0.8186     | 0.7352   |
| 8     | 0.9217     | 0.4700   | 0.8526     | 0.6563   | 0.8094     | 0.7906   |
| 9     | 0.9198     | 0.4801   | 0.8529     | 0.6723   | 0.8103     | 0.8292   |
| 10    | 0.9130     | 0.4919   | 0.8434     | 0.6939   | 0.8021     | 0.8736   |
| 11    | 0.9031     | 0.5066   | 0.8266     | 0.7173   | 0.7823     | 0.9207   |
| 12    | 0.8887     | 0.5231   | 0.8030     | 0.7379   | 0.7517     | 0.9450   |
| 13    | 0.8773     | 0.5508   | 0.7886     | 0.7814   | 0.7389     | 0.9758   |
| 14    | 0.8689     | 0.6466   | 0.7750     | 0.9366   | 0.7205     | 1.0914   |
| 15    | 0.8483     | 0.7470   | 0.7421     | 1.0809   | 0.6761     | 1.2154   |
| 16    | 0.8462     | 0.7314   | 0.7412     | 1.0568   | 0.6714     | 1.1852   |
| 17    | 0.8439     | 0.7346   | 0.7386     | 1.0655   | 0.6721     | 1.1842   |
| 18    | 0.8419     | 0.7212   | 0.7307     | 1.0388   | 0.6534     | 1.1499   |
| 19    | 0.8346     | 0.7391   | 0.7197     | 1.0673   | 0.6438     | 1.1787   |
| 20    | 0.8292     | 0.7351   | 0.7094     | 1.0599   | 0.6262     | 1.1735   |
| 21    | 0.8202     | 0.7442   | 0.6925     | 1.0677   | 0.6018     | 1.1903   |
| 22    | 0.8180     | 0.7361   | 0.6898     | 1.0536   | 0.5976     | 1.1735   |
| 23    | 0.8022     | 0.7523   | 0.6669     | 1.0705   | 0.5763     | 1.1943   |
| 24    | 0.7919     | 0.7523   | 0.6563     | 1.0619   | 0.5666     | 1.2004   |
| 25    | 0.7810     | 0.7420   | 0.6422     | 1.0453   | 0.5488     | 1.1808   |
| 26    | 0.7856     | 0.7130   | 0.6434     | 1.0099   | 0.5431     | 1.1377   |
| 27    | 0.7899     | 0.6891   | 0.6533     | 0.9926   | 0.5578     | 1.1282   |
| 28    | 0.7900     | 0.6630   | 0.6654     | 0.9655   | 0.5931     | 1.0612   |
| 29    | 0.7854     | 0.6594   | 0.6616     | 0.9628   | 0.5920     | 1.0690   |
| 30    | 0.7801     | 0.6534   | 0.6656     | 0.9714   | 0.5963     | 1.0895   |
| 31    | 0.6555     | 1.0205   | 0.5312     | 1.6291   | 0.4446     | 1.8246   |

# Plan for Next Week

- Finish the evaluation experiments
- Explore more ideas for designing an uncalibrated activation quantization technique working for DLMs